

Including Constraints

Objectives

After completing this lesson, you should be able to do the following:

- **Describe constraints**
- **Create and maintain constraints**

What are Constraints?

- Constraints enforce rules at the table level.
- Constraints prevent the deletion of a table if there are dependencies.
- The following constraint types are valid:
 - **NOT NULL**
 - **UNIQUE**
 - **PRIMARY KEY**
 - **FOREIGN KEY**

Constraint Guidelines

- Name a constraint.
- Create a constraint either:
 - **At the same time as the table is created, or**
 - **After the table has been created**
- Define a constraint at the **column or table level**.
- View a constraint in the data dictionary.

Types of Constraints

Constraint	Description
NOT NULL	Specifies that the column cannot contain a null value
UNIQUE	Specifies a column or combination of columns whose values must be unique for all rows in the table
PRIMARY KEY	Uniquely identifies each row of the table
FOREIGN KEY	Establishes and enforces a foreign key relationship between the column and a column of the referenced table

Defining Constraints

```
CREATE TABLE [schema.] table
    (column datatype [DEFAULT expr]
    [column_constraint],
    ...
    [table_constraint] [, ...]);
```

```
CREATE TABLE employees(
    employee_id  NUMBER(6),
    first_name   VARCHAR2(20),
    ...
    job_id       VARCHAR2(10) NOT NULL,
    CONSTRAINT emp_emp_id_pk
        PRIMARY KEY (EMPLOYEE_ID));
```

Defining Constraints

- Column constraint level

```
column [CONSTRAINT constraint_name] constraint_type,
```

- Table constraint level

```
column, ...  
    [CONSTRAINT constraint_name] constraint_type  
    (column, ...),
```

The NOT NULL Constraint

Ensures that null values are not permitted for the column:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	JOB_ID	SALARY	DEPARTMENT_ID
...						
176	Akhtar	Ali	24-MAR-98	SA_REP	8600	80
177	Ali	Zubari	23-APR-98	SA_REP	8400	80
178	Amjad	Ali	24-MAY-99	SA_REP	7000	
179	Fakhar Uz	Zaman	04-JAN-00	SA_REP	6200	80
180	Ibrar	Malik	24-JAN-98	SH_CLERK	3200	50
181	Danyal	Tahir	23-FEB-98	SH_CLERK	3100	50
182	Imran	shahid	21-JUN-99	SH_CLERK	2500	50
...						
107 rows selected.						

NOT NULL constraint
(No row can contain
a null value for
this column.)

**NOT NULL
constraint**

**Absence of NOT NULL
constraint**
(Any row can contain
null for this column.)

The NOT NULL Constraint

Is defined at the column level:

```
CREATE TABLE employees (  
    employee_id    NUMBER(6) ,  
    last_name      VARCHAR2(25) NOT NULL,  
    salary         NUMBER(8) ,  
    commission_pct NUMBER(2) ,  
    hire_date      DATE  
    CONSTRAINT  
emp_hire_date_nn  
                NOT NULL ,
```

← System
named

← User
named

The UNIQUE Constraint

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL
100	Salim	Mehmood	salim
101	Abdul	Rasheed	arasheed
102	Zia Ul	Hassan	ziahassan
103	S.M.	Ali	smali
104	Niaz	Hussain	niaz
105	Aziz	Khan	aziz
106	Yasir	Ali khan	yasir
107	Nazir	Janjua	nazir

UNIQUE constraint



...



INSERT INTO

301	Ahmad	Ali	ahmad
302	Ahmad	Fareed	ahmad



Allowed



Not allowed:
already exists

The UNIQUE Constraint

Defined at either the table level or the column level:

```
CREATE TABLE employees(  
    employee_id      NUMBER(6) ,  
    last_name        VARCHAR2(25)  
NOT NULL,  
    email            VARCHAR2(25) ,  
    salary           NUMBER(8) ,  
    commission_pct   NUMBER(2) ,  
    hire_date        DATE NOT NULL,  
    ...  
    CONSTRAINT emp_email_uk  
    UNIQUE(email)) ;
```

The PRIMARY KEY Constraint

DEPARTMENTS



PRIMARY KEY

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500

...

Not allowed
(Null value)



INSERT INTO

	Public Accounting		1400
50	Finance	124	1500

Not allowed
(50 already exists)



The PRIMARY KEY Constraint

Defined at either the table level or the column level:

```
CREATE TABLE departments(  
    department_id          NUMBER(4) ,  
    department_name        VARCHAR2(30)  
        CONSTRAINT dept_name_nn NOT NULL,  
    manager_id            NUMBER(6) ,  
    location_id            NUMBER(4) ,  
    CONSTRAINT dept_id_pk PRIMARY  
KEY(department_id));
```

The FOREIGN KEY Constraint

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500

**PRIMARY
KEY**



...



EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
200	Haroon	Javed	10
201	Toufiq	Ali	20
202	A.D.	Khan	20
114	Rehan	Akram	30
115	Rizwan	Shahid	30
116	Saad	Akhtar	30
119	Ghulam	Fareed	30

**FOREIGN
KEY**



...

INSERT INTO



305	Waris	Ali	9
306	Kareem	Khan	60

**Not allowed
(9 does not
exist)**



Allowed



[Example](#) link

The FOREIGN KEY Constraint

Defined at either the table level or the column level:

```
CREATE TABLE emp (
  employee_id      NUMBER(6) ,
  last_name        VARCHAR2(25)
NOT NULL,
  email            VARCHAR2(25) ,
  salary           NUMBER(8) ,
  commission_pct   NUMBER(2) ,
  hire_date        DATE NOT NULL,
  ...
  department_id    NUMBER(4) ,
  CONSTRAINT emp_dept_fk FOREIGN
KEY (department_id)
  REFERENCES
departments(department_id) ,
  CONSTRAINT emp_email_uk
UNIQUE(email)) ;
```

FOREIGN KEY Constraint Keywords

- **FOREIGN KEY:** Defines the column in the child table at the table constraint level
- **REFERENCES:** Identifies the table and column in the parent table

Adding a Constraint Syntax

Use the **ALTER TABLE** statement to:

- Add or drop a constraint, but not modify its structure
- Enable or disable constraints
- Add a **NOT NULL** constraint by using the **MODIFY** clause

```
ALTER TABLE table  
ADD [CONSTRAINT constraint] type (column);
```

Adding a Constraint

Add a FOREIGN KEY constraint to the EMPLOYEES table indicating that a manager must already exist as a valid employee in the EMPLOYEES table.

```
ALTER TABLE      employees
ADD CONSTRAINT    emp_manager_fk
    FOREIGN KEY (manager_id)
    REFERENCES employees (employee_id);
```

Table altered.

Dropping a Constraint

- Remove the manager constraint from the EMPLOYEES table.

```
ALTER TABLE      employees
DROP CONSTRAINT    emp_manager_fk;
Table altered.
```

Q/A